DEEP LEARNING · FINAL PROJECT

Quantum Dynamics: Predicting Qubit Evolution with Deep Learning

Sequence models for PXP quantum dynamics forecasting

Deep Learning course, Artificial Intelligence and Computer Science — University of Calabria, A.Y. 2025/2026

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Problem & Tasks

- **Goal:** exponentially costly $N = 10$ PXP quantum simulation $\rightarrow$ learn a fast data-driven surrogate ($D = 55$ observables, $T_{\text{out}} = 40$)
- **One leak-free pipeline, two model families:** RNN (M1 $\rightarrow$ M2), Transformer (M3 $\rightarrow$ M4) + Lab 6 Autoencoder pilot
- **Three core challenges** tackled alongside rigorous statistical validation (3 seeds + 5-Fold CV)

CHALLENGE 01

Input window

Minimize $T_{\text{in}} \in \{5, 10, 20, 40\}$ on strictly aligned targets

CHALLENGE 02

Super-resolution

Reconstruct native $\Delta t = 0.02$ s from $2\times$ coarser input

CHALLENGE 03

Interpretability

Explain physical dynamics via multi-head attention maps

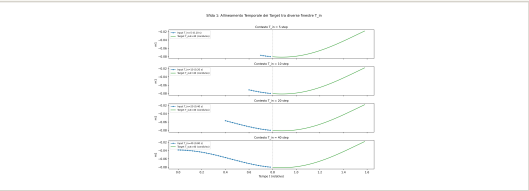
Dataset & Preprocessing

Data

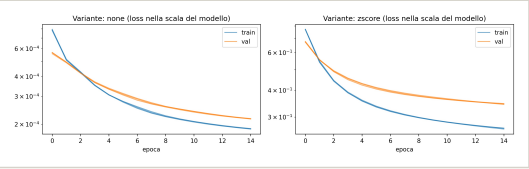
- PXP, $N = 10$ qubits, 400 trajectories, $\Delta t = 0.02$ s
- 55 features: 10 magnetizations $\langle \sigma_z^i \rangle + 45$ correlations $\langle \sigma_z^i \sigma_z^j \rangle$
- Bounded $[-1, 1]$ (0% violations) $\rightarrow$ $\tanh$ head, no normalization (`none`: $2.17\text{e-}4$ vs. Z-score: $2.32\text{e-}4$)

Setup

- Split per trajectory: $280/60/60$, `seed` $= 42$
- Windows: `short` $= 10$ / `long` $= 20$, `output` $= 40$, `stride` $= 10$
- Aligned targets (`align_t_in` $= 40$) for Challenge 1



Sliding-window extraction with strictly aligned future targets ($t_0 = 0.80$ s)



Normalization ablation: raw physical scale (`none` + $\tanh$) vs. Z-score

Model Roadmap

Lab 6 AE — Start-Small pilot ($55 \to 16 \to 55$) + Latent Transformer (-60.5% params)
Limit: AE reconstruction floor ($3.62e\text{-}4$) exceeds full-space accuracy → 55D kept

M1 — Direct multi-step LSTM/GRU encoder, single-shot Dense($40 \times 55, \tanh$)
Limit: compresses entire history into a single fixed state vector $\mathbf{h}_{T_{in}}$

M2 — Autoregressive LSTM/GRU seq2seq with $(\mathbf{h}, \mathbf{c})$ state transfer, 3-phase training
Limit: context stuck in 2 fixed vectors + recursive exposure bias

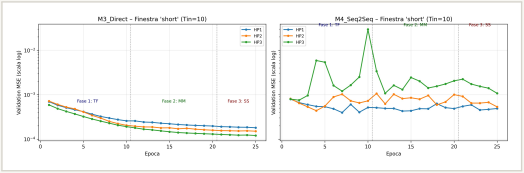
M3 — **Transformer Direct** (`GlobalSelfAttention` encoder + single-shot $\tanh$ head)
Best accuracy (Test MSE $= 1.21e\text{-}4$, zero drift); limit: no decoder cross-attention

M4 & SR — Full Encoder–Decoder (`CausalSelfAttn` + `CrossAttn`) & $2\times$ Super-Resolution
Extracts 40×10 attention maps & reconstructs native $\Delta t = 0.02$ s from $\Delta t = 0.04$ s

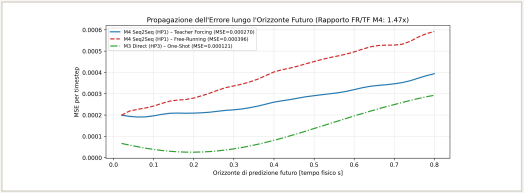
Training Curriculum

- Teacher Forcing (10) → Masked Modeling (10) → Scheduled Sampling (5) = 25 epochs
- **Direct models (M1, M3):** random 15% timestep masking ($p_{\text{mask}} = 0.15$) forces non-local context interpolation
- **Seq2Seq models (M2, M4):** two-pass Scheduled Sampling ($p : 0.0 \to 0.5$) mitigates open-loop recursive drift

10	10	5
Teacher Forcing	Masked Modeling	Scheduled Sampling



Validation MSE across the 3-phase curriculum (M3 Direct vs. M4 Seq2Seq)



Step-by-step error propagation: Teacher Forcing vs. Free-Running vs. One-Shot M3

Hyperparameter Tuning

- Full grid evaluation:
  3 configs (HP1, HP2, HP3) $\times$ 2 windows (`short`, `long`) $\times$ 4 architectures $= 24$ models
- Multi-seed stability (seeds $42, 43, 44$) + 5-Fold Trajectory Cross-Validation (340 trajectories) on top winners

M1 / M2 (RNN WINNER — HP1 / HP2)

`cell_type`	**LSTM**
`rnn_units`	**128**
`num_layers`	**1 (M1) / 2 (M2)**
`dropout`	**0.10**
`lr (AdamW + Cosine)`	**0.001**

M3 / M4 (TRANSFORMER WINNER — HP3 / HP1)

`d_model`	**64**
`num_heads`	**2 (M3) / 4 (M4)**
`num_layers (blocks)`	**1 (M3) / 2 (M4)**
`ff_dim`	$\mathbf{2 \cdot d_{model} = 128}$
`lr (AdamW + Cosine)`	**0.001**

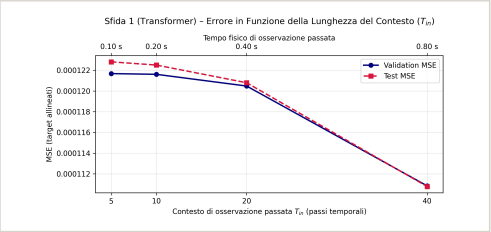
Test Set Performance & Statistical Validation

MODEL	3-SEED VAL MSE (SHORT)	5-FOLD CV MSE	TEST MSE (SHORT / LONG)	INFERENCE MODE / GAP
Persistence Baseline	$6.21\text{e-}4$	—	$6.23\text{e-}4$ / $6.21\text{e-}4$	n/a
M1 (LSTM direct, HP1)	$1.61\text{e-}4 \pm 1.3\text{e-}6$	$1.60\text{e-}4 \pm 2.1\text{e-}6$	$1.61\text{e-}4$ / $1.47\text{e-}4$	one-shot (no drift)
M2 (LSTM seq2seq, HP2)	$2.66\text{e-}4 \pm 5.0\text{e-}6$	—	$2.66\text{e-}4$ / $2.42\text{e-}4$	$2.66\text{e-}4$ (AR rollout)
M3 (Transformer direct, HP3)	$\mathbf{1.21\text{e-}4 \pm 1.0\text{e-}6}$	$\mathbf{1.20\text{e-}4 \pm 2.3\text{e-}6}$	$\mathbf{1.21\text{e-}4}$ / $\mathbf{1.21\text{e-}4}$	**one-shot (-25% vs. M1)**
M4 (Transformer seq2seq, HP1)	$4.02\text{e-}4 \pm 1.3\text{e-}5$	—	$3.96\text{e-}4$ / $3.38\text{e-}4$	TF $\mathbf{7.80\text{e-}5}$ → FR $3.96\text{e-}4$

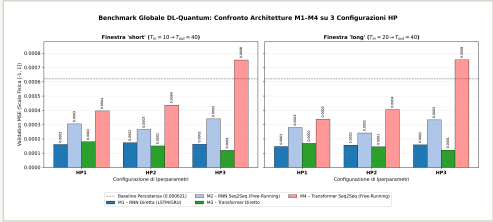
Key result: One-shot **M3** eliminates recursive exposure bias, reducing global MSE by $\mathbf{25}\%$ and single-qubit magnetization MSE by $\mathbf{50}\%$ ($3.60\text{e-}5$ vs. $7.20\text{e-}5$) over the best RNN.

Challenge 1: Minimal Input Window

- Evaluated across $T_{\text{in}} \in \{5, 10, 20, 40\}$ on **strictly aligned targets** (`align_t_in` $= 40$, same $5,580$ test segments)
- **M1 (LSTM):** improves from $T_{\text{in}} = 5$ $(1.72\text{e-}4)$ to $T_{\text{in}} = 20$ $(1.48\text{e-}4)$, then **saturates completely** at $T_{\text{in}} = 40$ $(1.48\text{e-}4)$
- **M3 (Transformer):** beats the best LSTM even at $T_{\text{in}} = 5$ $(1.23\text{e-}4)$ and scales to $\mathbf{1.11\text{e-}4}$ at $T_{\text{in}} = 40$ (-9%)



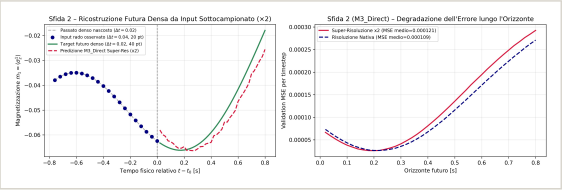
Aligned-target MSE scaling across $T_{in} \in \{5, 10, 20, 40\}$ for M3 Transformer



Validation MSE comparison across all four model families (short vs. long window)

Challenge 2: Super-Resolution

- **Test MSE 1.21e-4** (MAE 0.00788) from 2× coarser input (20 steps at $\Delta t_{\text{sub}} = 0.04$ s)
- **Matches standard M3** (1.21e-4) and sits within ΔMSE = 1.1e-5 of dense 39-step native M3 (1.10e-4)
- **No architecture change:** $f_{\text{Nyq}} = 12.5$ Hz $\gg f_{\text{scar}} \approx 0.7$ Hz allows exact PXP phase recovery



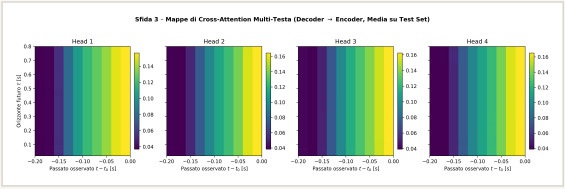
M3-SR reconstruction of a 2× subsampled test trajectory and horizon error profile

SUPER-RESOLUTION VS. NATIVE SAMPLING (ALIGNED TARGETS)

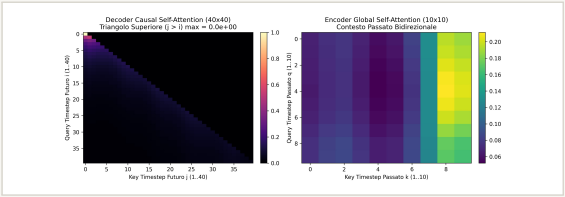
Setting	Δt_{in}	Test MSE	Test MAE
M3-SR (×2, 20 pts)	**0.04 s**	**1.21e-4**	**0.00788**
M3 Native (39 pts)	0.02 s	1.10e-4	0.00754
M3 Standard `short` (10 pts)	0.02 s	1.21e-4	0.00785

Challenge 3: Attention Maps

- **Heads 1 & 4 (Boundary Anchoring):** cross-attention concentrates on final input steps $t \in [-0.04, 0.00]$ s to lock initial state and slope
- **Heads 2 & 3 (Quantum-Scar Memory):** non-local attention bands across earlier lags track PXP revival frequency
- **Strict Causal Mask (40×40):** upper triangle $\max_{j>i} |\alpha_{ij}| = \mathbf{0.00}$ (zero future leakage)

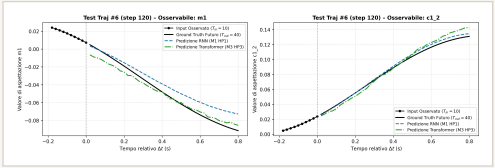


M4 decoder cross-attention across all 4 heads (40 future queries $\times$10 past keys)

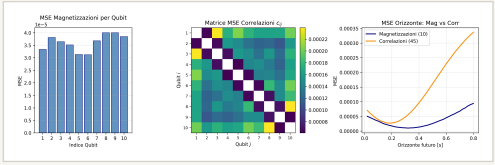


Decoder lower-triangular causal mask (40×40) and encoder bidirectional self-attention

Blind Test Forecast & Physical Analysis



Forecast comparison on blind test Trajectory #6 (m_1 and $c_{1,2}$), M1 RNN vs. M3 Transformer



Disaggregated physical error: per-qubit magnetization MSE, c_{ij} matrix, and horizon drift

- **M3 vs. M1 on Blind Test ($5,760$ windows in 1.72 s):** M3 tracks turning points of $m_1(t)$ and $c_{1,2}(t)$ with superior phase fidelity
- **Boundary vs. Bulk Qubits:** edge qubits (q_1, q_{10}) achieve lower MSE than central bulk qubits (q_5, q_6) due to open-chain boundary reflection
- **Physical Consistency:** 100% of test predictions satisfy two-body quantum positivity $(\min P(s_i, s_j) = \mathbf{0.1632} > \mathbf{0})$

Future Work

01

Longer horizon

Chain 40-step M3 blocks in block-autoregressive rollouts to extend $T_{\text{out}} \geq 160$ with $40\times$ fewer feedback steps

02

Coarser super-resolution

Test $4\times$ and $8\times$ subsampling limits ($\Delta t_{\text{sub}} \in \{0.08, 0.16\}$ s) to locate the empirical Nyquist breakdown of PXP scars

03

Larger qubit systems

Replace the fixed 55D projection with a spatial Graph-Attention encoder over the Rydberg chain to scale past $N = 10$

Thank you — Questions?

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University of Calabria — A.Y. 2025/2026 · https://github.com/Riccardo-27/DL-Quantum